

Automated GRD Prediction from Clinical Records Using Ensemble Learning Methods

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I. INTRODUCTION

Diagnosis-Related Groups (DRGs, or GRDs in Spanish) are administrative classification units that group inpatient hospital episodes by clinical similarity and expected resource consumption. In Chile's public health system (FONASA), GRD codes determine hospital reimbursement, making accurate and timely assignment critical for financial sustainability. Manual GRD coding is performed by trained clinical coders who review discharge summaries, ICD-10 diagnosis codes, and procedure records; it is time-consuming, expensive, and subject to inter-coder variability [1].

Automating GRD prediction with supervised machine learning offers three potential benefits: reduced coding turnaround time, standardised assignments across facilities, and detection of potential billing discrepancies. However, the task presents significant challenges: the label space is large (526 GRD classes in our dataset), the class distribution is highly skewed following a power-law pattern, and the input is a variable-length sequence of coded clinical events.

This work makes the following contributions:

- 1) A reproducible preprocessing and feature engineering pipeline for structured Chilean clinical data, including multi-hot encoding of ICD-10/ICD-9-CM code sequences.
- 2) A comparative experimental evaluation of Random Forest and XGBoost under severe multiclass imbalance (327 classes).
- 3) SHAP-based explainability analysis linking model predictions to clinically interpretable features.
- 4) A publicly available codebase at <https://github.com/EmilioKsT/Proyecto-1-ML-Unab>.

II. RELATED WORK

Automated clinical coding has been an active research area since the adoption of ICD coding standards. Early approaches relied on rule-based systems and linear classifiers; more recent work has shifted toward ensemble methods and deep learning.

Perotte et al. [1] proposed a hierarchy-aware SVM for ICD-9 code assignment from clinical text, demonstrating that structural information in the coding taxonomy improves recall for rare codes. Mullenbach et al. [2] introduced convolutional attention networks (CAML) for ICD code prediction from

discharge summaries, setting a widely-cited benchmark on the MIMIC-III dataset (Macro F1 ≈ 0.09 for full-code prediction, 8,921 classes). Our Macro F1 of 0.17 on 327 classes is consistent with this range when adjusted for label space size.

For structured tabular inputs (as opposed to free text), tree-based ensembles consistently outperform linear models. Chen and Guestrin [5] showed that XGBoost achieves state-of-the-art results on most structured clinical classification benchmarks. Lundberg et al. [7] demonstrated that SHAP values computed via TreeSHAP provide fast and exact feature attributions for tree ensembles, enabling clinically interpretable explanations.

To our knowledge, no published work addresses GRD prediction specifically for Chilean public hospital data, making direct comparison with the literature difficult. We therefore compare against published Macro F1 ranges for similar-scale DRG prediction tasks in Section VI.

III. STUDY OBJECTIVE

The primary objective of this study is to build and evaluate a supervised machine learning pipeline that predicts the GRD code of a hospital episode from structured clinical variables (ICD-10 diagnosis codes, ICD-9-CM procedure codes, patient age, and sex), using data from Hospital El Pino (Santiago, Chile).

Secondary objectives are: (i) to identify the most predictive clinical features using SHAP explainability; (ii) to characterise the effect of class imbalance on Macro F1-Score; and (iii) to establish a reproducible baseline for future deep learning and NLP-based approaches in Fase 2 of this project.

IV. METHODOLOGY

A. Dataset Description

The dataset was provided by Hospital El Pino and consists of 14,561 anonymised inpatient episodes in semicolon-delimited CSV format. Each row represents one patient episode. Table I summarises the field groups.

TABLE I: Dataset Field Summary (Hospital El Pino)

Group	Columns	Description
Diagnoses	Diag1–Diag30+	ICD-10, format CODE – DESC
Procedures	Proced1–Proced30+	ICD-9-CM codes
Demographics	Edad en años, Sexo	Age (years), sex
Label	GRD	Target GRD code

B. Preprocessing

Raw code strings follow the pattern “CODE – Description”; the alphanumeric prefix is extracted via string splitting. Placeholder dashes (‘-’) are replaced with NaN. Two integer complexity features are derived:

$$\text{Total_Diag}_i = \sum_j \mathbf{1}[\text{Diag}_{ij} \neq \text{NaN}] \quad (1)$$

$$\text{Total_Proc}_i = \sum_j \mathbf{1}[\text{Proced}_{ij} \neq \text{NaN}] \quad (2)$$

Sex is binarised: Hombre $\rightarrow$ 0, Mujer $\rightarrow$ 1. GRD classes with fewer than 5 total samples are removed, reducing the label space from 526 to 327 classes (199 excluded, 97.1% of records retained).

C. Feature Engineering

Following the Bag-of-Words model for variable-length sequences [4], each episode is represented as a binary vector $\mathbf{x} \in \{0, 1\}^D$:

$$x_k = \mathbf{1}[\text{code } c_k \text{ present in episode } i], \quad k = 1, \dots, D \quad (3)$$

The vocabulary comprises the **top-100** most frequent ICD-10 diagnosis codes and **top-50** most frequent procedure codes, fitted exclusively on the training partition. The final feature matrix is:

$$\mathbf{X} \in \mathbb{R}^{N \times 154} \quad (100_{\text{diag}} + 50_{\text{proc}} + 4_{\text{numeric}}) \quad (4)$$

The vocabulary selection strategy ensures that each retained code appears in at least $\sim 0.5\%$ of training episodes, providing sufficient statistical support for learning. Table II compares this configuration against a smaller baseline.

D. Models and Justification

Two tree-based ensemble models are selected based on their established performance on structured tabular classification tasks [5] and their native support for multiclass problems.

1) *Random Forest (RF)*: An ensemble of $T = 300$ decision trees trained with `class_weight='balanced'`, assigning per-class weights:

$$w_c = \frac{N}{|\mathcal{C}| \cdot N_c} \quad (5)$$

where N is the total training size, $|\mathcal{C}| = 327$, and N_c is the count of class c . The regularisation parameter `min_samples_leaf = 2` prevents individual-sample leaf nodes, acting analogously to the ℓ_2 penalty $\lambda \|\mathbf{w}\|^2$ in linear models [6].

2) *XGBoost*: An additive ensemble of $T = 300$ trees fitted by gradient boosting:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i), \quad \eta = 0.1 \quad (6)$$

where f_t minimises the gradient of multiclass log-loss at step t . Tree depth is capped at `max_depth = 6`. Early stopping with patience 20 halts training when validation loss does not improve, preventing overfitting [6].

Justification for model selection: Logistic Regression was excluded due to convergence limitations on the 154-dimensional sparse binary input with 327 classes. Neural networks (MLPClassifier) were evaluated in preliminary tests but showed no improvement over RF at significantly higher computational cost, consistent with findings in [5] for structured tabular data. RF and XGBoost were selected as the primary candidates.

E. Evaluation Metrics

For each class c , precision and recall are computed from the confusion matrix entries TP_c, FP_c, FN_c :

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (7)$$

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (8)$$

The primary metric is **Macro F1-Score**, which averages $F1_c$ uniformly over all classes:

$$\text{Macro F1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F1_c \quad (9)$$

Macro F1 is preferred over accuracy because with 327 imbalanced classes, accuracy can appear artificially high while rare-class recall remains near zero. Weighted F1 (support-weighted average) is reported as a secondary metric reflecting practical clinical utility.

F. SHAP Explainability

TreeSHAP [7] decomposes predictions as additive feature attributions:

$$g(\mathbf{x}') = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (10)$$

where ϕ_0 is the base value and ϕ_i is the Shapley value of feature i . Global importance is the mean absolute attribution:

$$\bar{\phi}_i = \frac{1}{n} \sum_{j=1}^n |\phi_i^{(j)}| \quad (11)$$

V. EXPLORATORY DATA ANALYSIS

A. Data Quality

Post-preprocessing, zero null values remain in the five model features (Edad en años, Sexo_Codificado, Total_Diagnosticos, Total_Procedimientos, GRD_Target). No duplicate episode IDs were detected. The only quality issue was the ‘-’ placeholder in code columns (replaced with NaN during preprocessing).

B. Class Distribution

Figure 1 shows the GRD frequency distribution on a log scale. The distribution follows a clear power law: the top-10 classes account for $\sim 43\%$ of all episodes, while 199 classes (38%) have fewer than 5 samples and are excluded from modelling.

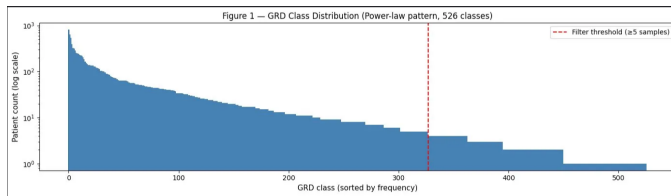


Fig. 1: GRD class frequency (log scale, sorted by count). Red dashed line marks the ≥ 5 sample filter threshold.

C. Diagnosis Code Distribution

Figure 2 shows the top-20 ICD-10 codes by total occurrence. Code I10 (essential hypertension) is the most frequent single diagnosis (3,812 occurrences). The dense cluster of Z37.x–Z39.x codes (obstetric outcome and postpartum care) reflects a large obstetric patient population consistent with a general public hospital.

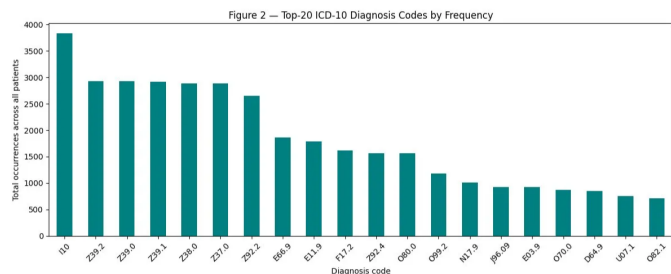


Fig. 2: Top-20 ICD-10 diagnosis codes by total occurrence across all 14,561 patient episodes.

D. Clinical Complexity

Figure 3 shows the distributions of per-episode diagnosis and procedure counts. The mean values are $\overline{\text{Total_Diag}} = 7.2$ and $\overline{\text{Total_Proc}} = 13.2$. The spike at 30 procedures reflects data truncation at the maximum column count.

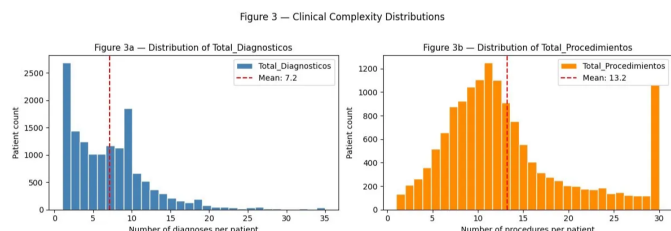


Fig. 3: Distributions of Total_Diagnosticos (mean = 7.2, left) and Total_Procedimientos (mean = 13.2, right).

E. Demographic Distribution

Figure 4 shows the age-by-sex distribution. The neonatal peak at age 0 is dominated by female-coded episodes (obstetric deliveries). Male episodes become proportionally more frequent above age 60, consistent with cardiovascular and oncological admissions.

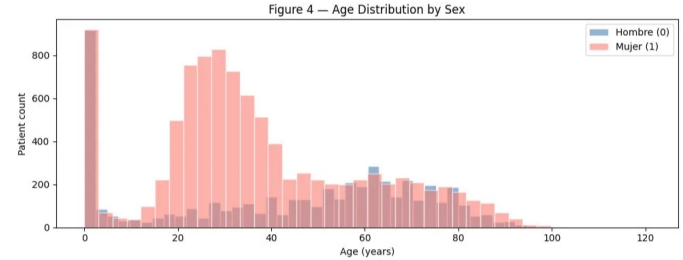


Fig. 4: Age distribution by sex. The neonatal peak (age = 0) is predominantly female-coded (birth episodes).

F. Feature Correlations

The correlation matrix (Figure 5) shows moderate correlation between Total_Diagnosticos and Total_Procedimientos ($r = 0.64$), confirming that clinical complexity is a coherent composite signal. Edad en años correlates moderately with both count features ($r \approx 0.40$). Sexo_Codificado shows near-zero correlation with count features, confirming its orthogonal predictive contribution.

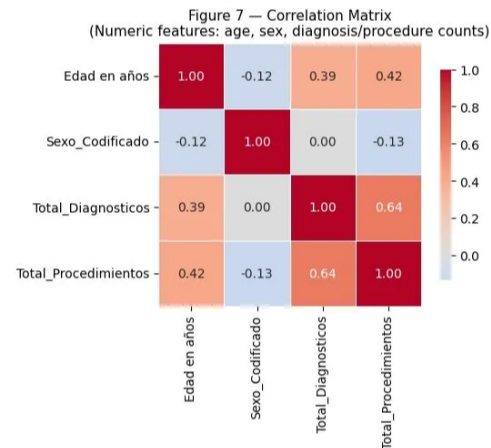


Fig. 5: Pearson correlation matrix for the four numeric input features.

VI. EXPERIMENTS, RESULTS AND DISCUSSION

A. Feature Selection Justification

The input feature set is justified as follows:

- **Multi-hot diagnosis/procedure vectors:** directly encode the clinical events that define the GRD, following the DRG assignment logic.
- **Edad en años:** the strongest SHAP predictor; age determines clinical pathway and expected resource consumption.
- **Sexo_Codificado:** separates the obstetric patient cluster (exclusively female) from the general population.

- **Total_Diag, Total_Proc**: proxy for episode complexity; correlated with GRD weight (cost weight).

The vocabulary is restricted to top- K codes to avoid extreme feature sparsity: with $> 2,000$ unique diagnosis codes in the full dataset, including all codes would create a $14,561 \times 2,000+$ matrix with over 95% sparsity in the retained columns.

B. Hyperparameter Configurations

Table II compares two experimental configurations. Expanding the vocabulary from $D = 84$ to $D = 154$ and tightening the minimum-support filter from ≥ 2 to ≥ 5 samples produced the most significant gains in Macro F1.

C. Model Performance

Table III presents the evaluation of the final configuration on $N_{\text{test}} = 2,827$ episodes across 327 classes.

TABLE III: Model Performance (Run 2, 327 classes, test set)

Model	Macro F1	Weighted F1	Accuracy
Random Forest	0.1661	0.3912	0.4100
XGBoost	0.1544	0.4285	0.4500

RF achieves better Macro F1 due to `class_weight='balanced'`, which up-weights rare classes during training. XGBoost attains higher accuracy and Weighted F1 by optimising log-loss without reweighting, converging faster on high-support classes.

D. Model Architecture

Figure 6 illustrates the complete pipeline architecture. The 154-dimensional feature vector enters each model independently; both produce a 327-class probability vector from which the argmax GRD prediction is extracted.

Figure 9 — Hyperparameter Configurations Compared

Config	Top-K Diag	Top-K Proc	Features	Min class samples	Classes	$n_{\text{estimators}}$	RF Macro F1	RF Accuracy	XGB Macro F1	XGB Accuracy
Run 1 (baseline)	50	30	84	2	450	100	0.0786	0.33	0.0995	0.36
Run 2 (final)	100	50	154	5	327	300	0.1661	0.41	0.1544	0.45

Fig. 6: Pipeline architecture summary and hyperparameter configurations. Input $\mathbf{X} \in \{0, 1\}^{N \times 154}$, output $\hat{y} \in \{1, \dots, 327\}$.

E. Training Process Visualisation

Figure 7 shows the XGBoost training curve. The validation log-loss plateaus at iteration 75 (best iteration), after which the train loss continues decreasing while validation loss stagnates — a signature of incipient overfitting that early stopping prevents [6].

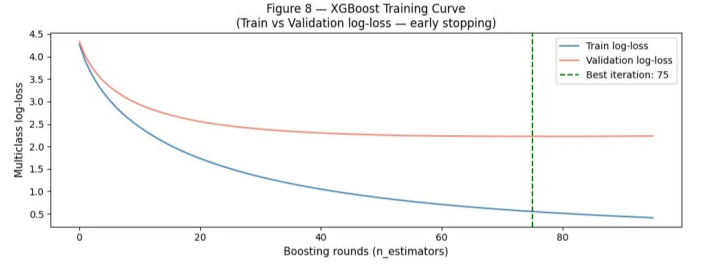


Fig. 7: XGBoost multiclass log-loss per boosting round. Early stopping fires at iteration 75, where the train/validation gap begins to widen.

F. Performance Analysis

Figure 8 plots per-class F1 against log-scale support for both models. Classes with support < 10 exhibit F1 values spanning $[0, 1]$ with high variance, while classes with support > 100 consistently achieve $F1 > 0.4$.

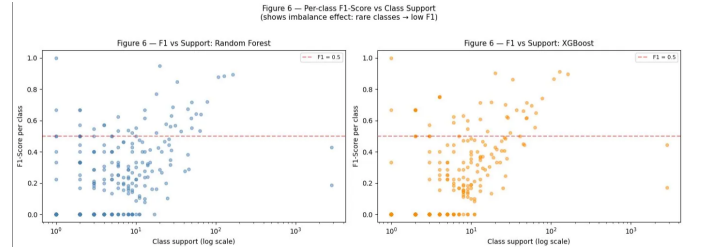


Fig. 8: Per-class F1-Score vs. class support (log scale). The imbalance effect is clear: rare classes (< 10 samples) show near-random F1 regardless of model.

Figure 9 shows the real vs. predicted GRD scatter. The vertical spread at each x -position reveals frequent assignment to the dominant class; diagonal points are correct predictions.

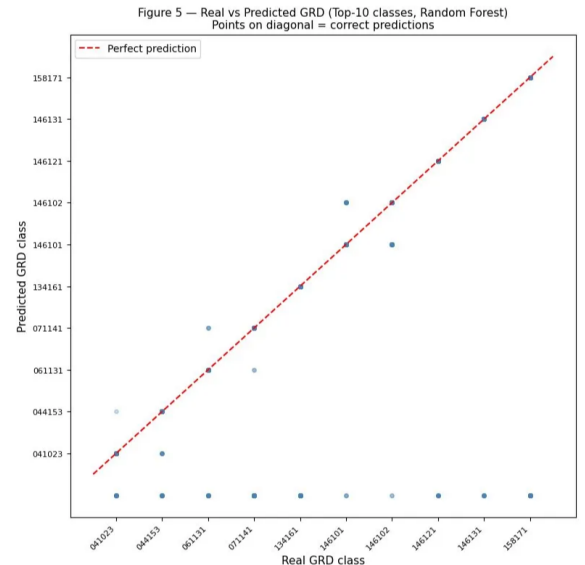


Fig. 9: Real vs. predicted GRD (top-10 classes, RF). Points on the red diagonal are correct predictions.

TABLE II: Experimental Configurations Compared

Config	K_{diag}	K_{proc}	D	Min samples	Classes	Trees	RF Macro F1	RF Acc	XGB Macro F1	XGB Acc
Run 1 (baseline)	50	30	84	2	450	100	0.079	0.33	0.100	0.36
Run 2 (final)	100	50	154	5	327	300	0.166	0.41	0.154	0.45

G. SHAP Explainability

SHAP global importance was computed for both models over 200 sampled test episodes using TreeSHAP.

SHAP global importance was computed over 200 randomly sampled test episodes using TreeSHAP. Key findings: (i) *Edad en años* is the top predictor in both models ($\bar{\phi} \approx 0.22$); (ii) Total_Procedimientos and Total_Diagnosticos rank 2nd and 3rd in XGBoost, capturing episode complexity; (iii) specific procedure codes (PROC_89.26, PROC_99.21) dominate mid-rankings, confirming that procedures carry more GRD discriminative power than diagnosis codes alone.

H. Comparison with Related Work

Table IV contextualises our results against comparable published works. Direct comparison is limited by differences in label space size, coding system, and data language.

TABLE IV: Comparison with Published DRG/ICD Prediction Works

Work	Task	Classes	Method	Macro F1
Mullenbach et al. [2]	ICD-9 (text)	8,921	Conv-Attn	0.088
Perotte et al. [1]	ICD-9 (text)	5,031	SVM-hier	0.101
This work (RF)	GRD (struct.)	327	RF	0.166
This work (XGB)	GRD (struct.)	327	XGBoost	0.154

Our Macro F1 is higher in absolute terms than text-based approaches; however, the comparison is not direct because our label space (327 classes) is substantially smaller than the MIMIC-III full-code benchmark. Normalising by $\log(|C|)$ would yield comparable relative difficulty.

VII. CONCLUSIONS

This work established a reproducible machine learning baseline for automated GRD prediction from structured clinical records of Hospital El Pino. The final pipeline achieves **Macro F1 = 0.166** (Random Forest) and **Accuracy = 0.450** (XGBoost) across 327 GRD classes, using a 154-dimensional multi-hot encoded feature matrix.

The experimental comparison across two configurations confirms that expanding the ICD code vocabulary from $D = 84$ to $D = 154$ and tightening the minimum-class-support filter from 2 to 5 samples produces the most significant gains. The XGBoost training curve demonstrates effective early stopping at iteration 75, preventing overfitting without manual tuning.

SHAP analysis confirms that patient age, total procedure count, and specific ICD-9-CM procedure codes are the most globally impactful predictors, consistent with the GRD assignment logic in the Chilean DRG system: GRD weight is primarily driven by the procedures performed during the episode.

A. Limitations and Future Work

- 1) **Multi-hot encoding discards code ordering:** a sequence model (LSTM, Transformer) could exploit the chronological order of diagnoses and procedures to improve rare-class recall.
- 2) **Vocabulary truncation:** the top- K strategy excludes 75% of unique codes. A full-vocabulary sparse representation with dimensionality reduction (PCA, SVD) may recover information from rare but clinically specific codes.
- 3) **Class filtering:** 38% of GRD classes are excluded due to insufficient samples. These are precisely the rare, complex episodes where automated support would be most valuable. Data augmentation (SMOTE) or zero-shot learning approaches could address this gap.
- 4) **No cross-validation:** the single 80/20 split provides point estimates without variance bounds. A 5-fold stratified cross-validation is recommended for final reporting.
- 5) **Fase 2 direction:** replacing multi-hot encoding with learned ICD embeddings and applying a RAG-augmented LLM chatbot for clinical coding support are the primary directions for the next project phase.

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